

Pubudu: Deep Learning Based Screening And Intervention of Dyslexia, Dysgraphia And Dyscalculia

Ruchira Kariyawasam¹, Madhuka Nadeeshani¹, Tuan Hamid¹, Inisha Subasinghe¹, Pradeepa Samarasinghe², Pasangi Ratnayake³

Faculty of Computing, Sri Lanka Institute of Information Technology

New Kandy Rd, Malabe, Sri Lanka

Abstract—Dyslexia, Dysgraphia and Dyscalculia are significant learning disabilities that affect around 10% of children in the world. Despite the advancement of technology literacy in the community, limited attention has been given for screening and intervention of these disabilities using mobile applications in Sri Lanka. In this research, one of the first deep learning and machine learning based mobile applications, named “Pubudu” was developed for screening and intervention of dyslexia, dysgraphia and dyscalculia supporting local languages. In “Pubudu” we have followed up clinical screening and diagnostic procedures recommended by health professionals for screening and intervention. The screening of dyslexia, letter dysgraphia and numeric dysgraphia was carried out using deep neural network and the screening for dyscalculia was carried out using machine learning techniques. Intervention techniques are implemented using gamified environments. System testing was carried out using 50 differently abled children and 50 typical children. With the initial dataset 88%, 58%, 99% screening accuracies are achieved in neural networks for letter dysgraphia, dyslexia and numeric dysgraphia screening while dyscalculia, whereas 90% accuracy was achieved for dyscalculia. Handwritten letters and numbers were fed as inputs to CNN model in letter dysgraphia and numeric dysgraphia while embedded audio clips of letter pronunciation were fed in to voice recognition CNN model in dyslexia. “Pubudu” shows significant potential for screening and intervention of dyslexia, dysgraphia and dyscalculia in local languages motivating children and interactively making them able and would be an enabling app for most of the underprivileged children in Sri Lanka.

Keywords—Dyslexia, Dysgraphia, Dyscalculia, Neural network, Machine learning, Mobile application

I. INTRODUCTION

According to the National Center for Learning Disabilities (NCLD) dyslexia, dysgraphia and dyscalculia are among the five most common types of learning disabilities around the world [1]. The Root causes for these disabilities are not yet defined clearly but most of them predict that these are due to certain neurological disorders. Dyslexia is a reading disability condition that is characterized by clinical symptoms such as difficulty with phonemic awareness, fluency, rate of reading, rhyming comprehension and written expression [2]. Approximately 15% of people in the world are dyslexic [3]. Dysgraphia is a disability condition in writing letters and numbers that is characterized by symptoms such as unclear, irregular handwriting often with different slants, difficult to write within the given scope and tend to write or copy things slowly [4]. About 5% - 33% students suffer from dysgraphia [5]. Dyscalculia is a mathematics disorder and it is estimated about 3% - 6% of children suffer from this learning disability [6]. It is characterized by difficulty in recognizing numbers, counting objects, classifying

mathematical signs (addition, subtraction, multiplication, and division), playing games that involve numbers and learning multiplication tables [7].

These learning disabilities that are commonly found in children have received less attention in the developing countries. Only few screening and intervention applications have been used deep learning and machine learning approaches and none of them address both screening and intervention of all three learning disabilities in one application. Through this research a mobile application is developed for screening and intervention of dyslexia, dysgraphia and dyscalculia of children in the age groups from six to seven. The primary objective of this application is to provide screening and intervention in the local language to enhance reading, writing and mathematical skills of those differently abled children and to uplift their educational standards.

II. LITERATURE REVIEW

In considering the existing applications in Sri Lanka, [8] was the first application to perform screening of dyslexia, dysgraphia and dyscalculia utilizing tablet computer game series having limitation to use the Kinect device. None of other researchers have focused on their attention screening all three learning disabilities in a single application.

A questionnaire-based screening test for both dyslexia and dysgraphic children was developed in [9] but they are not interactive for children and parent guidance is a must. A gamified environment for improving reading skills through playing action games was provided in [10] for preschoolers but the drawback was children have to practice these exercises daily to get an improvement. The purpose of study [11] is to conduct online screening of dyslexia followed by a questionnaire, but it is designed only for young adults. A study made by [12] has designed a user-friendly mobile application to assist children to recognize, memorize and repeat letters but this study does not support dyslexia screening. The study [13] was designed to assist people with dyslexia but less user friendliness and age group limitation was a major issue. [14] was a study done for diagnosis of dyslexia using computational analysis with 96% but it lacks to support language portability.

Out of the few screening apps for dysgraphia [15] was based on forward chaining with a good accuracy level, furthermore [16] was a study done for diagnosis of dysgraphia using handwriting classification methods to develop a statistical model for discriminating between **dysgraphic and proficient handwriting** with 90% accuracy but the limitation was less user-friendliness as a result parent guidance is a must. In study [17] two short and practical questionnaires were used to evaluate dysgraphia with 96%

accuracy but it doesn't support language portability and not interactive to children. A series of games were used in [18] for dysgraphia intervention but none of them was supporting language portability. A study [19] was done to identify lifts and stops in proficient and dysgraphic handwriting for roman writing for adults and children in tablet in less user-friendly manner.

Reviewing other research, [20] was designed as a standalone system for undergraduates not for school children, while [21] and [22] are computer-based screeners that are not interactive for children. For dyscalculia intervention [23] is a computer play tool and [24] is a web-based learning tool but neither of them supports local languages or a mobile application. Thus, to overcome the above given limitations the novel contributions of "Pubudu" are,

- 1) Screening and intervention of three learning disabilities dyslexia, dyscalculia, letter and numeric dysgraphia
- 2) Use of both Sinhala and English languages
- 3) Use of machine learning and neural networks for better accuracy in prediction and intervention.

III. METHODOLOGY

Four different strategies are adopted for the completion of the screening and intervention of dyslexia, dyscalculia and dysgraphia for letters and numbers. Fig. 1 and Fig. 2 show the overall activity diagram and system diagram for the system respectively.

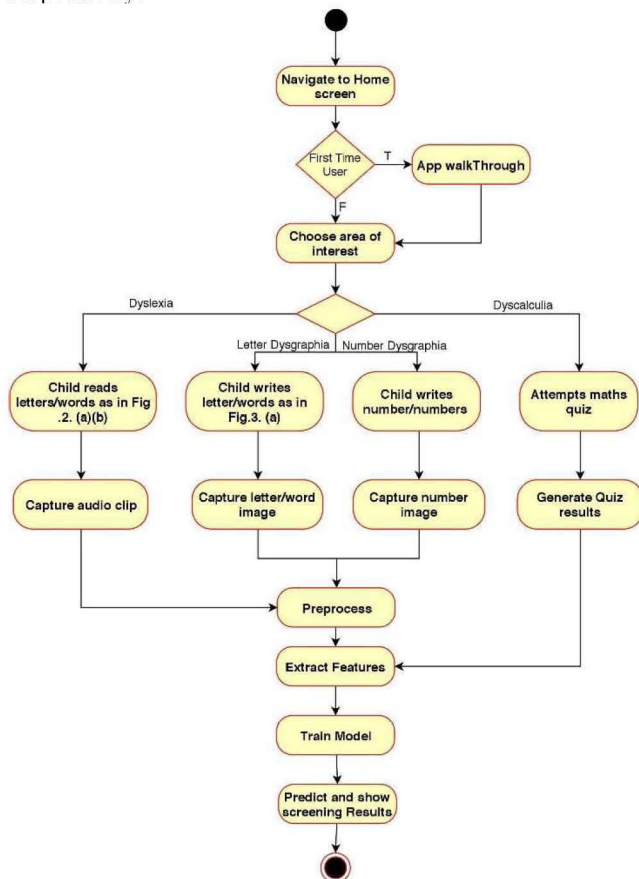


Fig. 1. System activity diagram

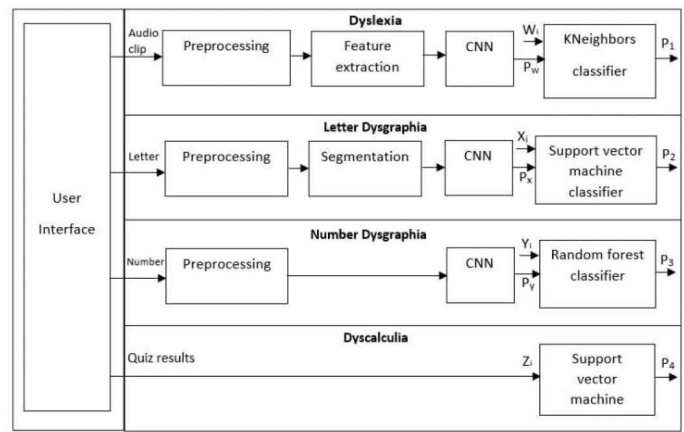


Fig. 2. System diagram

P_w – Success probability of pronounced letters/words

P_x – Success probability of written letters/words

P_y – Success probability of written number

W_i – {Total number of correctly pronounced letters/words, Total number of incorrectly pronounced letters/words, Total number of attempts, Total time taken to complete task}

X_i – {Total correct count of written letters/words, Total incorrect count of written letters/words, Total number of attempts, Total time taken to complete task, Total erased count}

Y_i – {Average time taken to write a number, Total erased count, Number of times that exceed boundary}

Z_i – {Stage1 score, Stage1 time, Stage2 score, Stage2 time, Stage3 score, Stage3 time}

P_1 – Probability of predicting dyslexia

P_2 – Probability of predicting letter dysgraphia

P_3 – Probability of predicting numeric dysgraphia

P_4 – Probability of predicting dyscalculia

CNN - Convolutional Neural Network

A. Screening and intervention of dyslexia in sinhala language



(c) Dyslexia intervention training

(d) Dyslexia intervention exercise

Fig. 3. Sample dyslexia interfaces

In dyslexia screening, a series of pronunciation tests are carried out to differentiate dyslexic and non-dyslexic children. As the first step, a single Sinhala letter is displayed as shown in Fig. 3.a and gradually the difficulty level was increased up to two letter words as shown in Fig. 3.b.

The CNN model is trained utilizing data gathered from 2000 non dyslexic school children within the age group from six to seven from different regions of the country to obtain an unbiased output. Voice recognition CNN model with 11 sequential layers was implemented comprising three conv2D layers, three dropout layers, three dense layers, one flatten layer, one max pooling layer and finally with a dense layer consisting Softmax activation function. Each conv2D layer comprises of (2*2) kernel size and relu activation function with 32, 48 and 120 number of filters. Maxpooling2D layer with (2*2) pool size. Two dense layers each with 128 and 64 units with relu as activation function. Each dropout layer containing 0.25 and 0.4 rates.

Since audio clips were gathered in their natural environment, preprocessing was done to remove noise. A fixed size 1D array audio clip vector was prepared using zero padding technique and fed to CNN. Feature extraction was done using Mel-Frequency Cepstral Coefficient (MFCC). Softmax yielded good accuracy in comparison with LeakyRelu, furthermore, the number of epochs was increased to 150 to gain a high accuracy level. Adadelata was used as the optimizer function. Repeatedly data were collected from both dyslexic and non-dyslexic children and they were fed in to KNN. Five parameters of KNN used are success probability of pronounced letters/words, total number of correctly pronounced letters/words, total number of incorrectly pronounced letters/words, total number of attempts and total time taken to complete the task. By utilizing those parameters it will predict whether user is having dyslexia or not.

If the child is predicted to be dyslexic a set of intervention procedures are proposed. Results are fed back into the model for higher accuracy. In intervention of dyslexia a multisensory environment was created with Cocos2dx game engine. In dyslexia intervention, children are given a training based on recalling words through pictures as shown in Fig. 3.c. Then they are tested in two stages. In the first stage users are asked to select corresponding letter in a balloon and secondly, users are asked to pronounce the word displayed as shown in Fig. 3.d. If she or he pronounced the word correctly that balloon will burst with a congratulating screen. Competency level was measured based on total time taken, total number of attempts taken in interventions. If intervention mechanisms were satisfactory then users will be navigated back to screening mechanisms if not users were retained in intervention stage to practice more.

B. Screening and intervention of letter dysgraphia

In dysgraphia screening, children were asked to write a single Sinhala letter with three attempts as shown in Fig. 4.a. After that child are made to write two Sinhala letters with similar number of attempts.

5000 images that were collected from non-dysgraphic children from age six to seven were used to CNN build the model. CNN model used for handwriting recognition consists of 3 conv2D layers, 3 maxpooling layers, 1flatten layer and 2 dense layers. Each conv2D layer contains 64,128 and 256 filters with (3, 3) kernel size and with 1 strides for each layer.

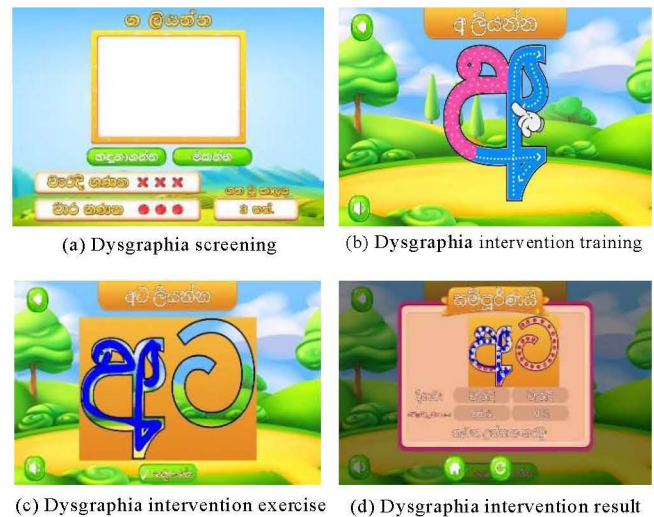


Fig. 4. Sample Letter Dysgraphia Interfaces

Each maxpooling2D layer with pool size 2 and with 2 strides. Dense layers with 1024 and 6 units. CNN model was utilized in order to identify whether the written letters of children were correct or not.

Images were converted to binary images through the binarization process and were preprocessed to eliminate noise. Preprocessed and normalized images are then separated to individual letters through segmentation process. CNN with 8 layers is used where Multi-class Cross Entropy is the loss function.

Six inputs were fed to SVM namely, success probability of written letters/words, total correct count of written letters/words, total incorrect count of written letters/words, total number of attempts, total time taken to complete task and total erased count in screening mechanism from both dysgraphic and non-dysgraphic children. By utilizing those it will predict whether users are having dysgraphia or not using SVM algorithm.

If the child was predicted to be dysgraphic she/he is navigated to intervention mechanisms. Initially they are trained on how to write letters in proceeding path with precise start and endpoints with animations in a defined scope as shown in Fig. 4.b. In latter stages users are allowed to write letters independently within given scope. In deciding the level of competency as shown in Fig. 4.c. users are allowed to write two letters in given scope and their total number of nodes covered and proceeding path of letters are taken into consideration. The summary of intervention results are displayed to the user as shown in Fig.4.d and if intervention mechanisms are satisfactory then users will be rescreened.

C. Screening and intervention of dyscalculia

The process to screen Dyscalculia involves a series of **mathematical tests which tests a student's mathematical ability**. Following are some of the skills used to measure Dyscalculia:

- 1) Dot Counting - Dot enumeration tasks, which are used to measure numerical capacity, subjects are required to count the number of dots displayed, usually less than 10 dots.
- 2) Number Comparison - Subjects are asked to select the numerically larger of two numbers. Subjects with

difficulty in recognizing numerosity may have failed in connecting numerals and their meaning.

- 3) Addition - Subjects are differentiated by the speed with a normal student retrieving the answer from memory versus counting using fingers.

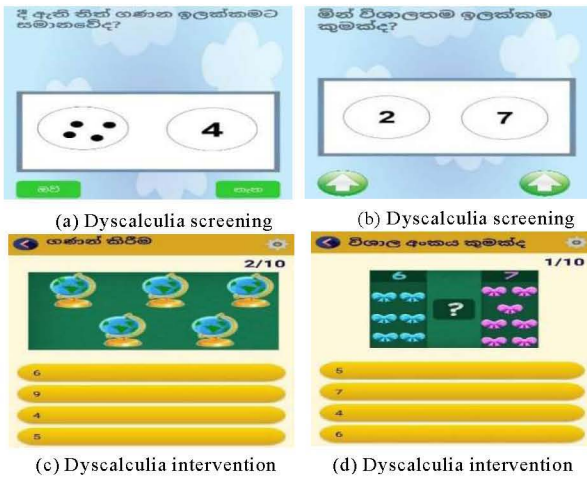


Fig. 5. Sample dyscalculia interfaces

The dyscalculia screening tests in “Pubudu” are split into the above three stages. The first stage checks the counting ability as shown in Fig.5.a and the second stage checks the number comparison ability as shown in Fig.5.b In the final stage, basic arithmetic addition is checked. Finally, the average time taken per question and overall score of each stage are the parameters considered when screening for dyscalculia and the model is built using SVM algorithm. Upon starting the screening process, the user will be taken through 3 stages containing 10 questions related to mathematical skills. The questions to be displayed are taken from the quiz bank.

Based on the results of this screening process, if a user is identified as having dyscalculia, the user will be directed to the intervention activities which teach math problems in a graphical way. An intervention provides training for counting as shown in Fig.5.c, training in number comparison as shown in Fig.5.d and training in arithmetic addition.

D. Screening and intervention of numeric dysgraphia

For screening of numeric dysgraphia numbers, initially a writing canvas is provided as shown in Fig. 6. a. Then the user is asked to write numbers from 0 to 9, each number twice. This is an unguided writing in which a child has more freedom to use the given space. To screen numeric dysgraphia (ND) following parameters are taken [23]; completion time, accuracy, number of times out of boundary and number of corrections. The accuracy of the number is predicted using a pre trained model. This model uses the MNIST handwritten digits dataset which contains a CNN [24]. The CNN is made up with 19 layers and 20 epochs were used to obtain a high accuracy from the CNN, which is 99.2%. The numbers model was then converted to a Tensor Flow Lite file.

The evaluation of the handwriting happens after the writing test is completed. average time taken to write a number, total erased count, number of times that exceed boundary and success probability of written numbers as parameters a Random Forest Classifier is built. The model will predict the probability of having numeric dysgraphia

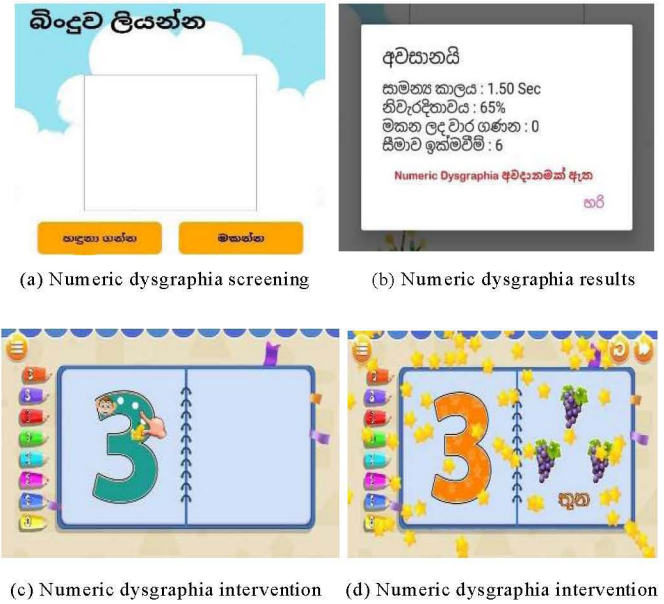


Fig. 6. Sample numeric dysgraphia interfaces

In training the model, a total of 100 children, 50 pre-diagnosed with dysgraphia and 50 typical are taken as subjects, these subjects are in the age groups from six to seven and studying in special education unit at Senehasa Education Resource Research & Information Centre and regular schools.

If a user is found ND positive, he/she will be notified as shown in Fig. 6.b and will be directed to the intervention process. There, he/she is prompted to trace the numbers which were previously written incorrectly and trains the child as shown in Fig. 6.c and Fig. 6.d The interventions game is designed in consultation with medical professionals in a way that it develops observation and motor skills of the kids.

IV. RESULTS

“Pubudu” was tested among 50 children having learning disabilities, studying in special units in schools around different regions of Sri Lanka and in Senehasa Education Resource Research & Information Centre. Accurate results were obtained when pre-diagnosed children with dyslexia, dysgraphia and dyscalculia were screened using “Pubudu”.

A. Dyslexia screening

1000 audio clips for each letter with 0.8 second duration, collected from non-dyslexic school children were fed to CNN model as training data. CNN will recognize whether user pronounce letters are correct or not. KNN will predict whether user is having dyslexia or not with above mentioned five parameters collected from both dyslexic and non-dyslexic children. With the CNN 85% of training accuracy and 65% of testing accuracy was achieved in screening of dyslexia. Although a lower average precision of 58% was obtained for the “Pubudu” application in Sinhala language, an average accuracy around 94% was obtained with English language due to a large dataset.

Screening was conducted by participation of 50 students where 20 students were pre-diagnosed with dyslexia. Once screened through “Pubudu” 13 were correctly identified with dyslexia giving 65% level of accuracy in screening.

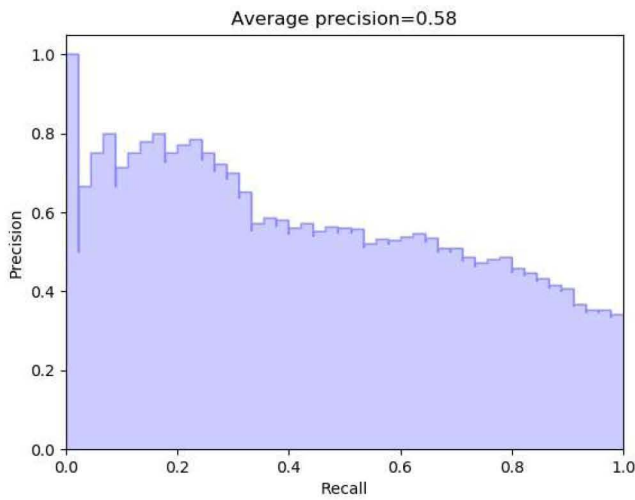


Fig. 7. Precision recall curve for dyslexia screening

B. Letter dysgraphia screening

For each letter, 5000 images were collected from nondysgraphic school children and then images were subjected to normalization to gain a fixed size of 50*50 pixels and words were separated to individual letters through the segmentation process. Finally, pre-processed and normalized images were fed into CNN model as training data to identify whether user written letters are correct or not. 95% training accuracy and 85% test accuracy was achieved by the neural network used for screening of letter dysgraphia. SVM will predict whether user is having dysgraphia or not with above mentioned six parameters gathered from both dysgraphic and non dysgraphic children.

CNN used for screening dysgraphia for letters reached a high accuracy over neural network used for screening of dyslexia due to the enriched database.

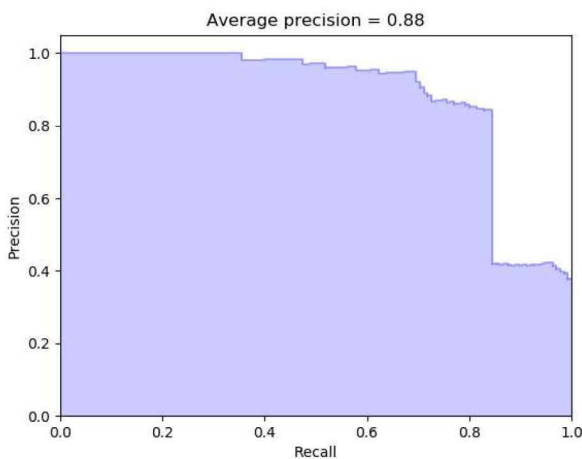


Fig. 8. Precision recall curve for letter dysgraphia screening

Out of 50 subjects used for screening of dysgraphia 20 were pre-diagnosed with dysgraphia. The screening process through “Pubudu” resulted in identifying 17 subjects with letter dysgraphia giving an overall accuracy around 85%.

C. Dyscalculia screening

For the SVM classifier where; True positive- Dyscalculia positive and detected, True negative- Dyscalculia negative

and not detected, False positive- Dyscalculia negative but detected, False Negative- Dyscalculia positive but not detected showed an accuracy of 0.92 and average precision of 0.94 as shown in Fig. 9.

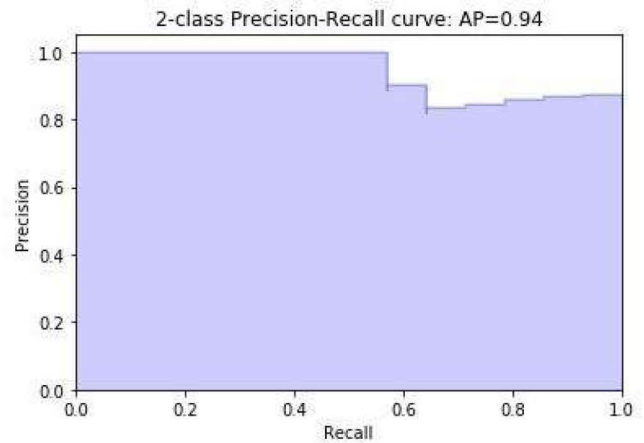


Fig. 9. Precision recall curve for dyscalculia screening

A set of fifty subjects were used for screening of dyscalculia, where 20 were pre-diagnosed with dyscalculia. The screening process through ‘Pubudu’ resulted in identifying 18 subjects with dyscalculia which gave an accuracy of 90%.

D. Numeric dysgraphia screening

The model used to check correctness of the hand-written numbers used the MNIST handwritten digits dataset which achieved a high testing and training accuracy of 99.2%.

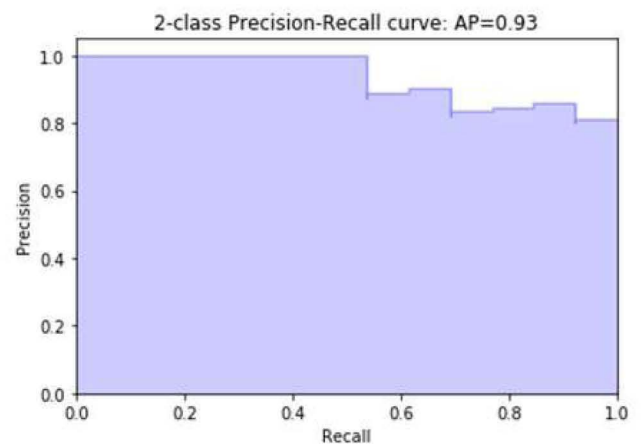


Fig. 10. Precision recall curve for numeric dysgraphia screening

For the Random Forest classifier which was used in screening ND where; True positive- ND positive and detected, True negative- ND negative and not detected, False positive- ND negative but detected, False Negative ND positive but not detected showed an accuracy of 0.88 and average precision of 0.93 as shown in Fig. 10.

20 subjects were pre-diagnosed with dysgraphia from a set of 50 subjects. Among them, 18 subjects were identified as dysgraphic for numbers through the screening process, which showed an accuracy of 0.9.

V. CONCLUSION

As per knowledge “Pubudu” is the first app in Sri Lanka to address learning disabilities dyslexia, dysgraphia and dyscalculia. Due to highly precise screening mechanisms employed, average results yielded for dyslexia, dysgraphia and dyscalculia reflect the success of this research. In order to increase accuracy, we plan to enrich our model with more data. The intervention of these children is still under progress as according to the recommendation of health professionals, children should be given at least six months period training through the application.

Though “Pubudu” supports android platform now, in future it will be upgraded to support other operating systems. Further “Pubudu” will be extended to support other local languages such as Tamil with the support of health ministry we plan to use “Pubudu” throughout the country making Sri Lanka a healthier country.

ACKNOWLEDGMENT

We are sincerely thankful to Col. Prasanna Hettiarachchi and Dr. Kamalani Wanigasinghe at Senhasa Education Resource Research and Information Centre (SERRIC) for their constant support and guidance throughout this research project. We are also extremely grateful to all the school children and their families who contributed in numerous ways to make this project a success.

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